AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 10 · WORKBOOK

Human Classifier Simulation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected classifier-procedure record.

Write a contract, construct both paths, repair a missing stop and distinguish validation from evaluation.

Keep one habit: Preserve the original input, rule version, first trace, repair and remaining unknowns.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Features and Distractors

Rows Columns and Data Types

Trace Tables

W1 · State the complete Tower1 input contract

> The accepted range is a declared rule, not a list of past fitting examples.

Reference, question and your record

- Tower1 reference for W1–W3: height is present, unit cm, whole number 1–10 inclusive. Check in that order. On a failed check record original input, Tower1, no prediction and the reason; stop that record. On acceptance predict tall at height >=6, otherwise short, and record original input, Tower1 and prediction. The threshold is supplied for this exercise, not fitted in this week.

- Write the complete Tower1 **input contract**. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted **input** does not have to be one of the examples used to fit the threshold.

- The threshold is given in this week. In your last explanation distinguish the accepted-input contract from whatever examples a future fitting exercise might use; no previous fitting record is needed to apply Tower1 here.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Write both paths and the output record

> A partner should not need to invent an instruction.

Your evidence record

- Tower1 reference for W1–W3: height is present, unit cm, whole number 1–10 inclusive. Check in that order. On a failed check record original input, Tower1, no prediction and the reason; stop that record. On acceptance predict tall at height >=6, otherwise short, and record original input, Tower1 and prediction. The threshold is supplied for this exercise, not fitted in this week.

- Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Trace the boundary and repair a missing stop

> Preserve the faulty path and its correct final outcome.

Your evidence record

- Tower1 reference for W1–W3: height is present, unit cm, whole number 1–10 inclusive. Check in that order. On a failed check record original input, Tower1, no prediction and the reason; stop that record. On acceptance predict tall at height >=6, otherwise short, and record original input, Tower1 and prediction. The threshold is supplied for this exercise, not fitted in this week.

- Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records **no prediction** for a blank, then continues and treats the blank as zero to report short.

  Identify the missing instruction and the proper final outcome.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Distinguish a valid Ribbon1 run from correctness

> Use the separate reference supplied here; do not invent a comparison label.

Your evidence record

- Separate Ribbon1 reference for W5: present whole-number ribbon length 1–12 inclusive with unit cm. Reject a failed contract check with no prediction and a reason, then stop. Accepted values predict long at >=8, short otherwise. Keep original input and Ribbon1 in each record. No comparison labels are supplied. This reference makes W5 self-contained even if you skip the optional full ribbon exercise.

- Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown.

  Explain why changing a blank to 1 cm just to produce a category would be a mistake.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Changed independent case

> Keep one first attempt before hints or feedback.

Read and explain independently

A **written rule** selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Construct both outcome paths for Constructing a **classifier**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Recommend a book only if genre is science AND pages<=150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A human **classifier** uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what **evidence** is missing in the second case.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 10 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Recommend a book only if genre is science AND pages<=150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Extra practice · Construct Ribbon1

REFERENCE · SECTION 15

Extra practice · Construct Ribbon1

> This is a separate optional construction task, not a copy of selected W4.

The ribbon contract, rule and trace request

- Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

  Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

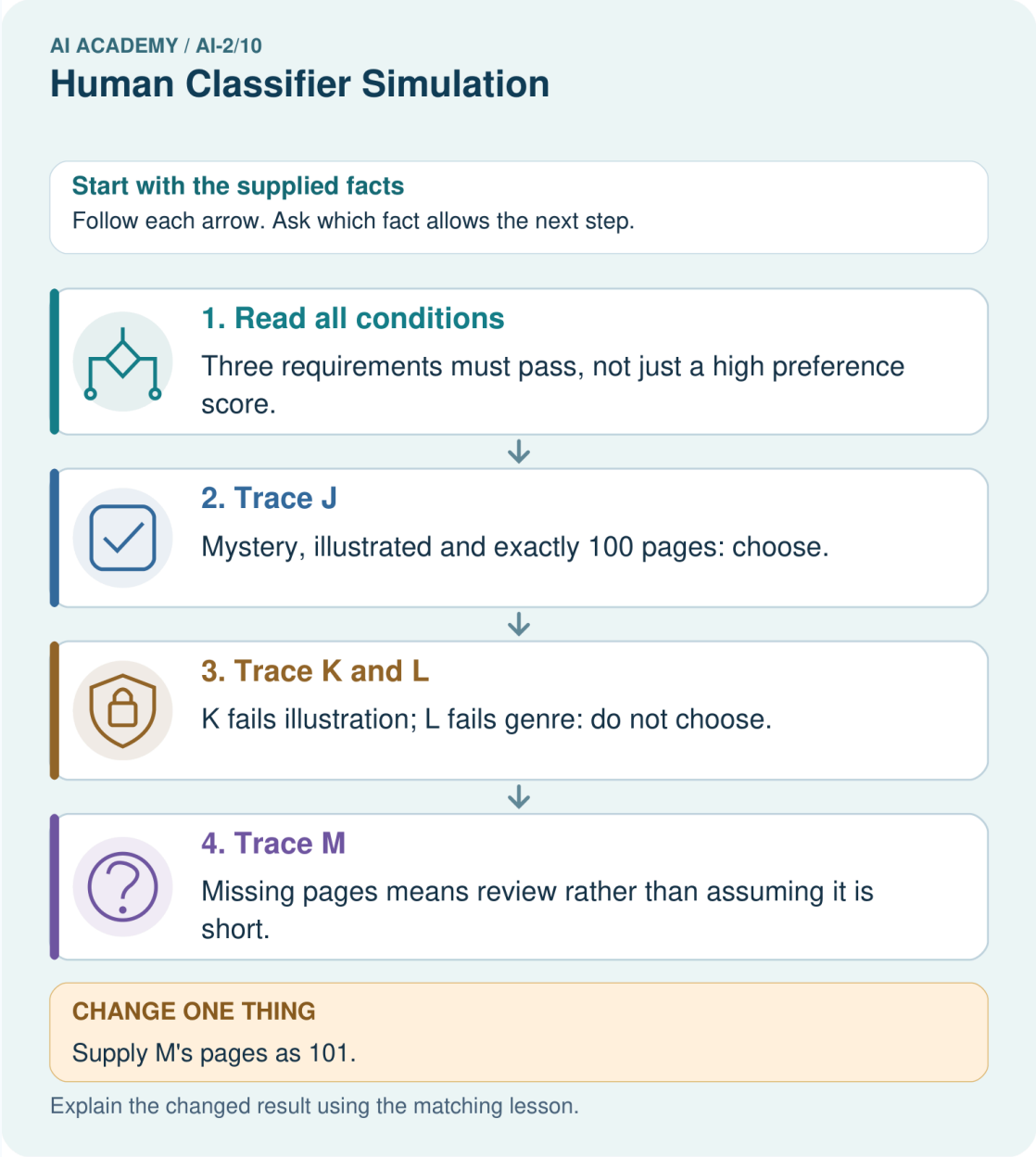
> Keep the explanation and the support you used.

Use this when helpful

A fictional librarian wants illustrated mystery books with at most 100 pages.

Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why are J and M initially treated differently even though both are illustrated mysteries?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional librarian wants illustrated mystery books with at most 100 pages.
- Which labelled step matches this detail? Missing pages means review rather than assuming it is short.
- 1. Read all conditions
- 2. Trace M

- 3. Trace J
- 4. Trace K and L
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Trace M Missing pages means review rather than assuming it is short. This human simulation lets students inspect every decision without needing a trained model.

Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

Changed-case question

- Changed-case practice: Supply M's pages as 101. Which conclusion is supported by the supplied facts?
- 1. M is now rejected rather than awaiting page review.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: M is now rejected rather than awaiting page review. The required field is known but fails the maximum-length condition.

Repair and save your work

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the **classifier** always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

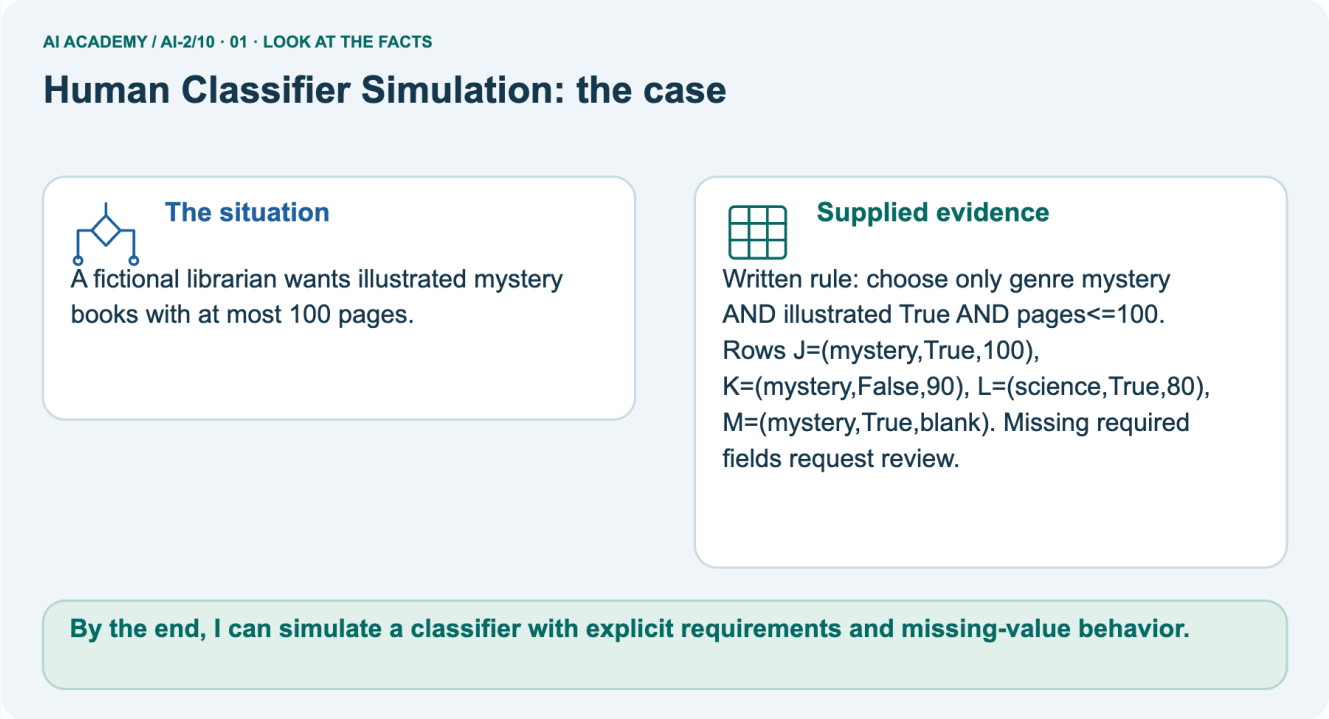
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Human Classifier Simulation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

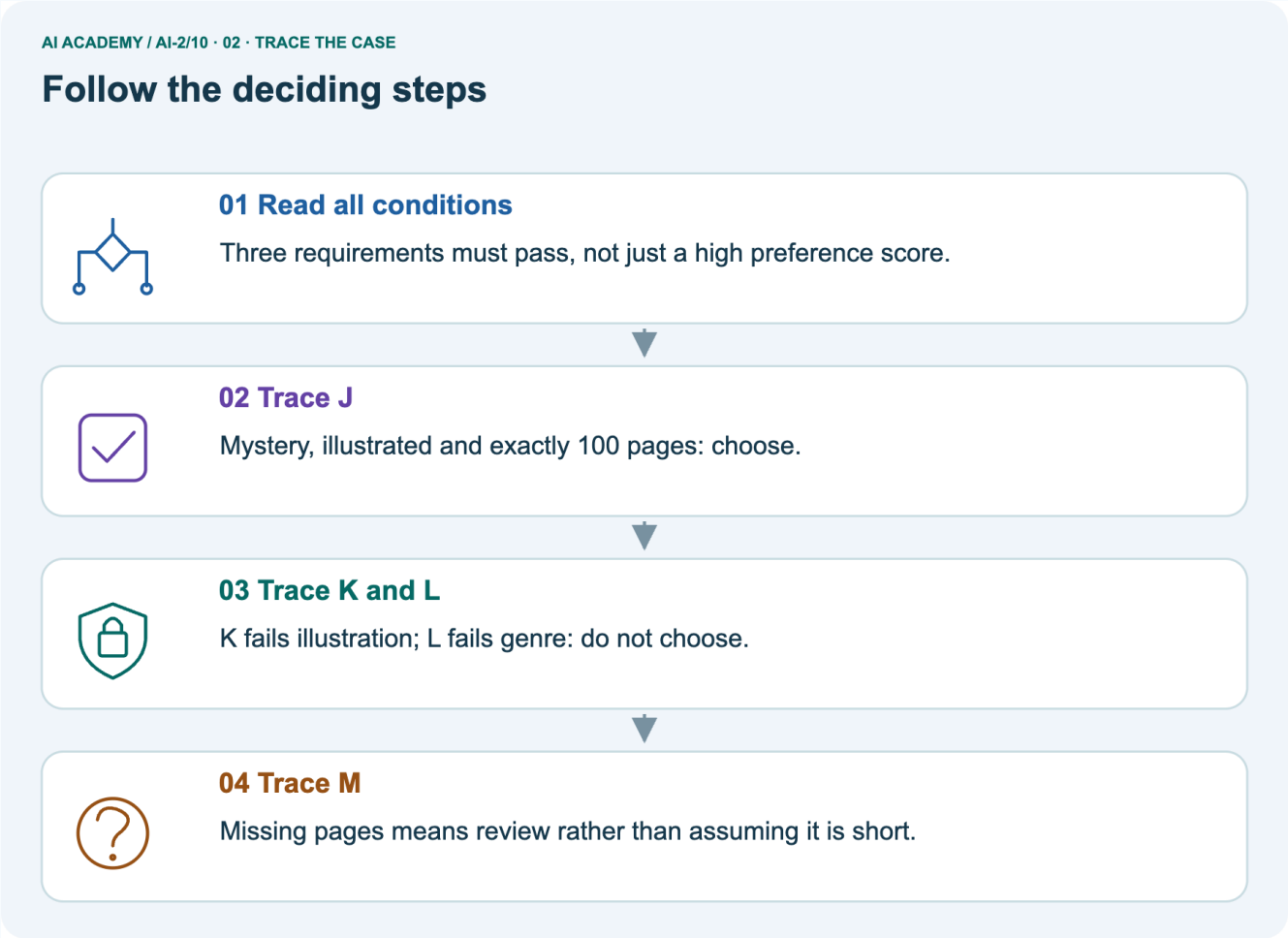
A fictional librarian wants illustrated mystery books with at most 100 pages. Supplied facts:
Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read all conditions: Three requirements must pass, not just a high preference score. → Trace J: Mystery, illustrated and exactly 100 pages: choose. → Trace K and L: K fails illustration; L fails genre: do not choose. → Trace M: Missing pages means review rather than assuming it is short.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three records through the Tower1 classifier

AI ACADEMY / AI-2/10 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

Follow the labelled picture with the matching explanation. Open the vector diagram

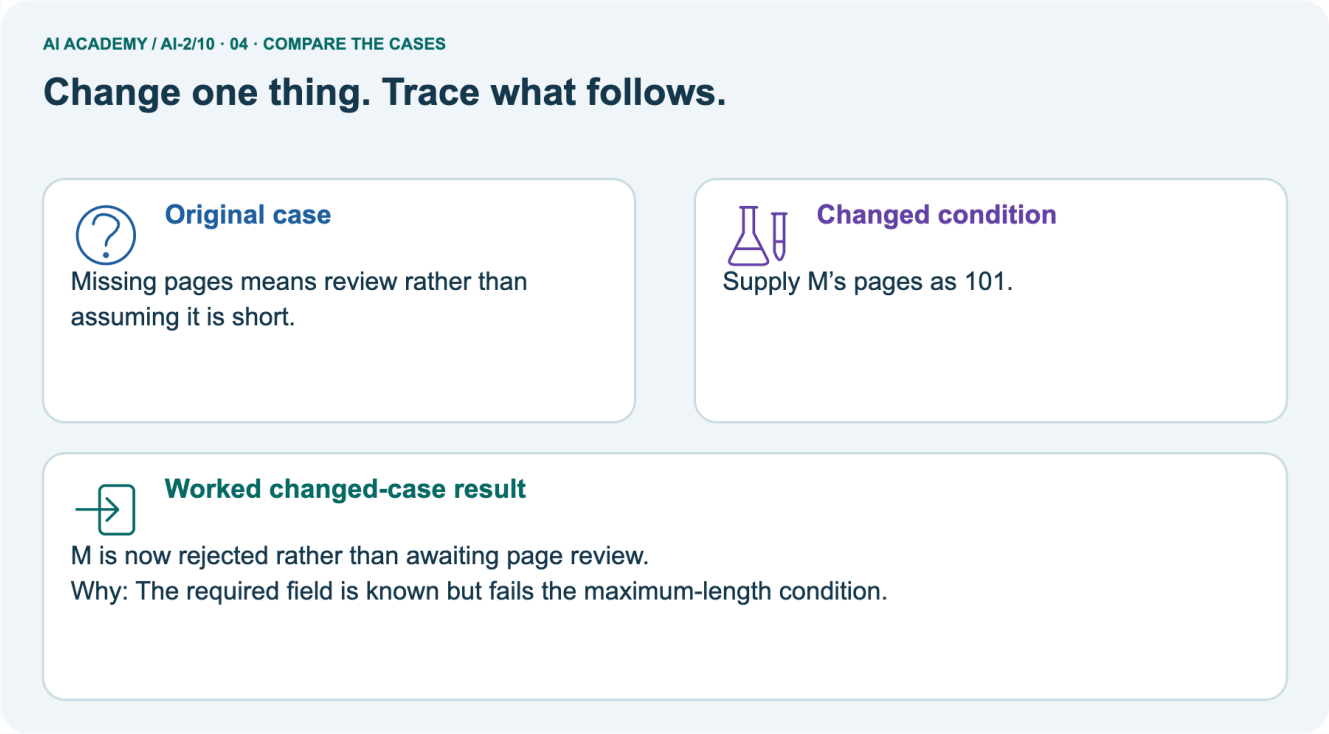
Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

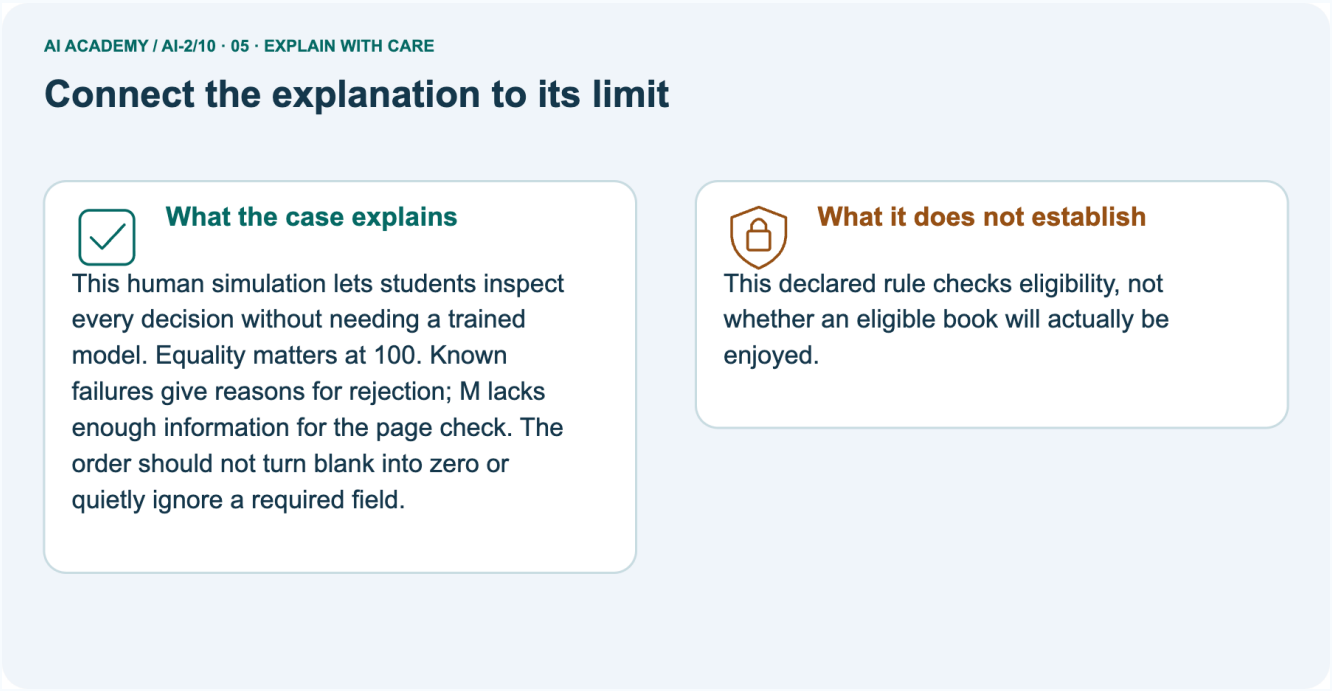
Original case: Missing pages means review rather than assuming it is short. Changed condition: Supply M's pages as 101. Expected result: M is now rejected rather than awaiting page review.

Reason: The required field is known but fails the maximum-length condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

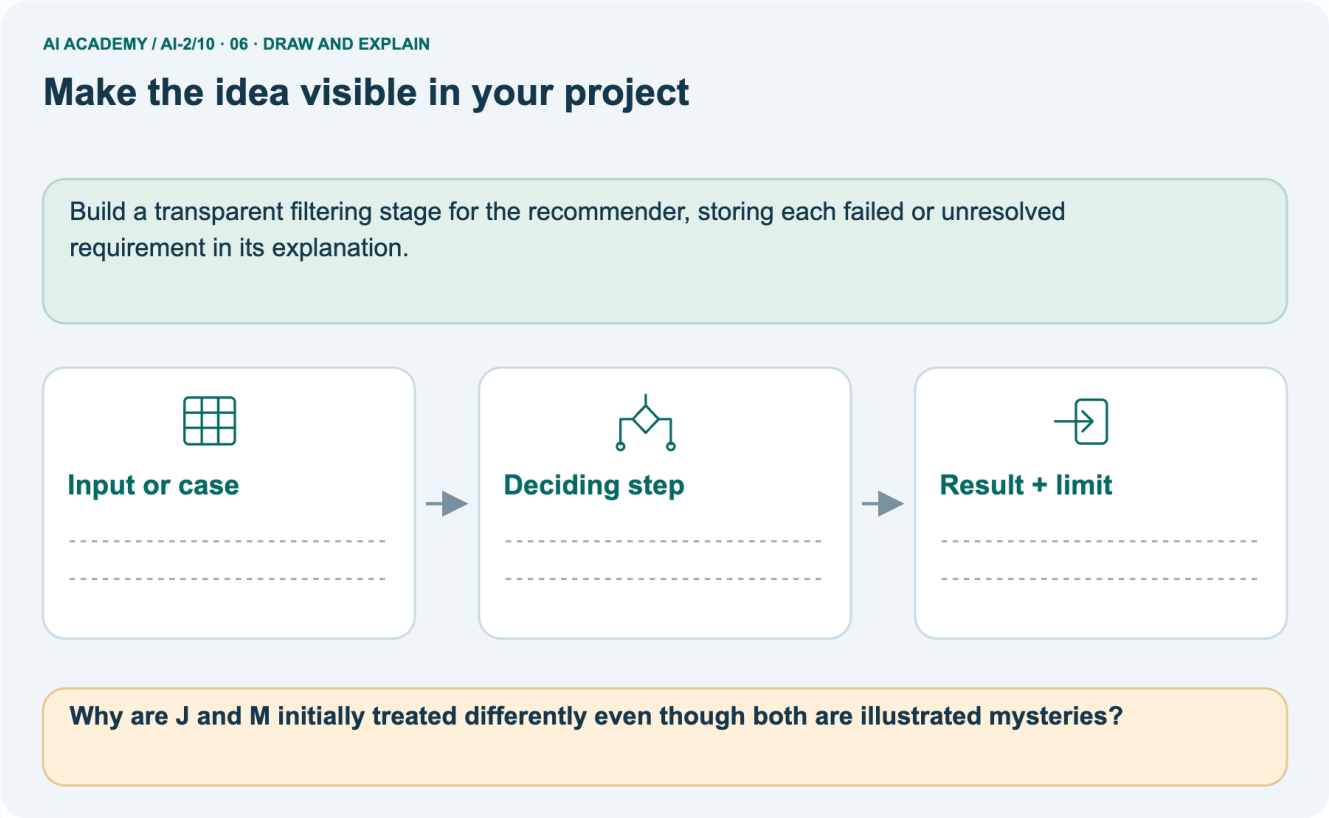
This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check.

The order should not turn blank into zero or quietly ignore a required field. Limit: This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation. Draw your **input**, deciding step, result and limit.

Supported question: Why are J and M initially treated differently even though both are illustrated mysteries?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Human Classifier Simulation: the case

02. Follow the deciding steps

03. Three records through the Tower1 classifier

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.